

ADVYTH VAMAN AKALANKAM

CSE Student @ Amrita Vishwa Vidyapeetham, Amritapuri

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Profile

Computer Science student focused on software engineering, data science, and applied AI. Experienced in building machine learning models, statistical data pipelines, and full-stack analytical applications using Python and modern ML frameworks.

Education

Amrita Vishwa Vidyapeetham

B.Tech in Computer Science and Engineering – CGPA: 8.8

FIITJEE Junior College

Class 12 – 97.2%

Dr KKR's Gowtham Concept School

Class 10 – CGPA: 10.0

Aug. 2023 – Present

Amritapuri, Kerala

Jun. 2021 – May 2023

Hyderabad, Telangana

Jun. 2020 – May 2021

Hyderabad, Telangana

Technical Skills

Languages: Python, Java, C++

Machine Learning & AI: Machine Learning, Deep Learning, Scikit-learn, HuggingFace, PyTorch, OpenCV, MediaPipe

Data Science: Pandas, NumPy, SciPy, Data Processing, Statistical Analysis, Data Visualization

Frameworks & APIs: Django, Flask, FastAPI, REST APIs, Streamlit, Plotly

Databases & Tools: PostgreSQL, MySQL, MongoDB, SQLite, Git, Linux, Vercel, Render

Experience

ACM Student Chapter, Amritapuri

SIG AI Lead & Member

Sep. 2023 – Present

Amritapuri, Kerala

- Led and organized 6+ AI/ML workshops for 150+ students, increasing learning output by 40%.
- Designed structured AI learning paths and mentored juniors, resulting in 10+ completed individual projects.
- Orchestrated AI induction for 200+ students, inspiring interest and boosting AI awareness in the incoming batch by nearly 60%.

Projects

Dynamic Risk Models (Portfolio Optimization) | Python, NumPy, SciPy, Streamlit

- Built a quantitative portfolio optimization dashboard implementing Markowitz, CVaR, and Black-Litterman models with out-of-sample backtesting.
- Applied hierarchical clustering to identify correlation structures and evaluate diversification and tail-risk behavior.

MarketPulse (Financial Analytics Dashboard) | Python, FastAPI, Pandas, Plotly, Streamlit

- Developed a financial analytics platform processing data from 2,300+ NSE-listed securities using automated data pipelines.
- Built a modular FastAPI backend and interactive Plotly dashboards for market analysis, price trends, trading volume, RSI, and MACD.

Smart Expense Analyzer | Flask, React, SQLite, Pandas, Scikit-learn

- Developed a full-stack expense analytics platform with REST APIs, JWT authentication, and interactive dashboards for transaction analysis.
- Built Pandas-based preprocessing pipelines and implemented Isolation Forest for machine learning anomaly detection on transaction data.

Community Outreach

Student Social Responsibility

- Participated in a Student Social Responsibility initiative focused on guiding students in exploring diverse career opportunities and making informed choices.
- Conducted awareness sessions on the ethical use of AI, promoting responsible and thoughtful technology usage.

Prizes & Publications

- Publication – *Intelligent Accident Rate & Fatality Analysis using OpenStreetMaps*, GRENZE Journal (2024).
- 2nd Place, InnovateK12 Hackathon (2024) – Developed an AI-based predictive analytics solution within 24 hours.